



**MICHIGAN DEPARTMENT OF ENVIRONMENTAL QUALITY
ENVIRONMENTAL LABORATORY**

P.O. Box 30270
Lansing, MI 48909
TEL: (517) 335-9800
FAX: (517) 335-9600

30 October 2014

Work Order: 1410104

Price: \$195.00

Chris Little

MDEQ-WRD-GRAND RAPIDS

350 Ottawa Ave NW, Unit 10

Grand Rapids, MI 49503-2341

RE: CADILLAC WWTP

I certify that the analyses performed by the MDEQ Environmental Laboratory were conducted by methods approved by the U.S. Environmental Protection Agency and other appropriate regulatory agencies.

Sincerely,

George Krisztian
Laboratory Director



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MDEQ-WRD-GRAND RAPIDS
350 Ottawa Ave NW, Unit 10
Grand Rapids MI, 49503-2341

Project: CADILLAC WWTP
Site Code: MI0020257
Project Manager: Chris Little

Reported:
10/30/2014

Analytical Report for Samples

Sample ID	Laboratory ID	Matrix	Date Sampled	Date Received	Qualifier
CD-001A	1410104-01	Waste Water	10/15/2014	10/15/2014	
CD-UP	1410104-03	Waste Water	10/14/2014	10/15/2014	
CD-001A-F	1410104-04	Waste Water	10/15/2014	10/15/2014	

Notes and Definitions

X3 Spike recovery is not applicable due to large target analyte concentration in the source sample.
ND Indicates compound analyzed for but not detected
RL Reporting Limit
NA Not Applicable

Case Narrative

Sample 1410104-02 field ID "CD-001A-1 was subcontracted for Available Cyanide analysis.


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Client ID: CD-001A

Lab ID: 1410104-01

CAS #	Analyte	Result	RL	Units	Dilution	Analyzed Date	QC Batch	Method	Qualifier
Inorganics-General Chemistry									
	Alkalinity-Total	130	20	mg/L	1	10/22/14	B4J2201	310.2	
7664-41-7	Ammonia	0.31	0.01	mg/L	1	10/22/14	B4J2212	350.1	
7723-14-0	Total Phosphorus	0.15	0.010	mg/L	1	10/27/14	B4J2706	365.4	
TSS	Total Suspended Solids	7	4	mg/L	1	10/21/14	B4J2109	2540 D	
Inorganics-Metals									
7440-70-2	Calcium	69	1.0	mg/L	1	10/28/14	B4J1704	6010/200.7	
7440-50-8	Copper	7.3	1.0	ug/L	1	10/28/14	B4J1704	6020/200.8	
	Hardness - Calculated	250	4.6	mg/L	1	10/28/14	[CALC]	2340 B	
7439-95-4	Magnesium	18	0.5	mg/L	1	10/28/14	B4J1704	6010/200.7	

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Client ID: CD-UP

Lab ID: 1410104-03

CAS #	Analyte	Result	RL	Units	Dilution	Analyzed Date	QC Batch	Method	Qualifier
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Inorganics-General Chemistry

	Alkalinity-Total	70	20	mg/L	1	10/22/14	B4J2201	310.2	
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Inorganics-Metals

7440-70-2	Calcium	22	1.0	mg/L	1	10/28/14	B4J1704	6010/200.7	
	Hardness - Calculated	75	4.6	mg/L	1	10/28/14	[CALC]	2340 B	
7439-95-4	Magnesium	4.9	0.5	mg/L	1	10/28/14	B4J1704	6010/200.7	

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Client ID: CD-001A-F

Lab ID: 1410104-04

CAS #	Analyte	Result	RL	Units	Dilution	Analyzed Date	QC Batch	Method	Qualifier
Inorganics-General Chemistry									
7664-41-7	Ammonia	1.4	0.01	mg/L	1	10/22/14	B4J2212	350.1	
7723-14-0	Total Phosphorus	0.15	0.010	mg/L	1	10/27/14	B4J2706	365.4	
TSS	Total Suspended Solids	6	4	mg/L	1	10/21/14	B4J2109	2540 D	

MICHIGAN DEPARTMENT OF ENVIRONMENTAL QUALITY

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ENVIRONMENTAL LABORATORY - ANALYSIS REQUEST SHEET

Lab Work Order Number

Project Name

Matrix

1416104

Cadillac WWTP

WASTE WATER

Site Code/Project Number

AY

CC Email 1 to send additional reports to

Project TAT Days

Sample Collector

MI0020257

15

Chris C. Little

Dept-Division-District

Index

CC Email 2 to send additional reports to

Project Due Date

Sample Collector Phone

MDEQ-WRD-PSM-GR

63206

cell 616-550-7668

State Project Manager

PCA 42202
42203

CC Email 3 to send additional reports to

Accept Analysis
hold time codes: Yes/No

Contract Firm

Chris C. Little

Project

Overflow Lab Choice 1

Contract Firm Primary Contact

State Project Manager Email

littlec@michigan.gov

481000

Overflow Lab Choice 2

Primary Contact Phone

State Project Manager Phone

cell 616-550-7668

Phase
00

Lab Use Only	Field Sample Identification	Collection Date	Collection Time	Container Count	Comments
1	01 CD-001A	10/15/14	1010		
2	02 CD-001A-1	10/14/14	1015		
3	03 CD-UP	10/14/14	1100		
4	04 CD-001A-F	10/15/14	0700		
5					
6					
7					
8					
9					
10					

MA - TOTAL METALS		GENERAL CHEMISTRY	
Copper - Cu	1 2 3 4 5 6 7 8 9 10	GCN - General Cyanide	
Hardness - Ca, Mg	1 2 3 4 5 6 7 8 9 10	Available Cyanide - CN	1 2 3 4 5 6 7 8 9 10
Shared bottle for line 1		GN - General Chemistry Neutral	
		Suspended Solids - SS	1 2 3 4 5 6 7 8 9 10
		MN - Inorganic Matrix Neutral	
		Total Alkalinity	1 2 3 4 5 6 7 8 9 10
		GA - General Chemistry Acidic	
		Ammonia - NH ₃	1 2 3 4 5 6 7 8 9 10
		Total Phosphorus - TP	1 2 3 4 5 6 7 8 9 10

Chain of Custody	Relinquished by	Received By	Date / Time
	Print Name & Org. Chris C. Little - MDEQ/WRD/PSM/GR	Melissa Smith	10/15/14 1431
	Signature: <i>Chris C. Little</i>	<i>Melissa Smith</i>	
	Print Name & Org.		
	Signature:		
	Print Name & Org.		
	Signature:		